0695. Max Area of Island

Medium | Category: Graphs | LeetCode: https://leetcode.com/problems/max-area

Problem

- Grid of 0/1. Return the largest island area (number of cells), or 0.

Intuition

- Number of Islands, but the flood fill returns how many cells it sank.

Brute Force

- BFS per island with a visited matrix: same complexity.

Optimal Approach

- Area(r, c): out of bounds or 0 → 0; else set 0 and return 1 + area of 4 neighbours.
- Answer = max over all cells.

Worked Example

```
0 1 1
1 1 0     cells (0,1),(0,2),(1,0),(1,1) connected -> area 4
0 0 1     cell (2,2) alone -> area 1
answer 4
```

Java Solution

```java
public class MaxAreaOfIsland {
    public int maxAreaOfIsland(int[][] grid) {
        int best = 0;
        for (int r = 0; r < grid.length; r++)
            for (int c = 0; c < grid[0].length; c++) best = Math.max(best, area(grid, r, c));
        return best;
    }

    private int area(int[][] g, int r, int c) {
        if (r < 0 || c < 0 || r >= g.length || c >= g[0].length || g[r][c] == 0) return 0;
        g[r][c] = 0;
        return 1 + area(g, r + 1, c) + area(g, r - 1, c) + area(g, r, c + 1) + area(g, r, c - 1);
    }
}
```

Complexity

- Time: O(m n). Space: O(m n) recursion worst case.

Pitfalls

- Mark before recursing so a cell is not counted twice.